2023 F=ma Exam Review

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- Average velocity $= \Delta x/t$. Initial position was 2.25 rotations ago. Two rotations ago it was in the same spot at the bottom; another fourth rotation ago it would be at left side of circle, so average velocity would be leftward and downward.

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- The work the person does $= mgh + W_f$, and $W_f = \mu mg\cos\theta \cdot \dfrac{h}{\sin\theta}$.

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- Look at the answer options—the height is $\ll$ hypotenuse, so use small angle approximations and relate the triangles of tension's components and the physical lengths triangle (l, r, h).

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- "Rest frame" means $v_{cm} = 0$, so total momentum $= 0$, so $m_a v_a = m_b v_b$ so $v_a/v_b = m_b/m_a = 1/10$.

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- Use the range formula $R = \dfrac{v^2 \sin 2\theta}{g}$.

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- Combine range and height formulas to get that $H = R$ when $\tan\theta = 4$.
- Max height happens when $\theta = 90$ and $H = v^2/2g$ which $= h/\sin^2(\arctan 4)$.

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- This one tests your physics sense. For the box to lift, it will reach $N = 0$, which means somewhere along the way friction (μN) will be not enough to stop sliding, so it will always slide first (before $N = 0$).

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- $v_{\text{orb}} = \sqrt{GM/r}$, and since v appears constant, $M \propto r$.

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- Since it is a small angle pendulum, use SHM equation $\theta(t) = \theta \cos \omega t$.
- Solve for t when $\theta(t) = \theta \cos \omega t = -\theta/2$.
- Make sure to recognize that is half the time between bounces.

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- It describes the picture for your imagination. Mass attached to end of string to form a small angle pendulum. It has v_x and v_y.
- $v_x = 0$ and $v_y = 0$ at top of trajectory so $(0, 0)$ should be on graph.
- Count how many times it should hit zero: two times per period for v_x, four times for v_y, so they should cross the y and x axis 2 and 4 times respectively.

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- Not fully sure about this one, but: $\sum F = 0$ so v_{cm} = constant and $\Delta p = 0$ and both ring and mass have mass of m so they are moving at $v/2$ relative to each other, then angular momentum conservation confirms r is at least $R/2$ but geometry limits it to max $R/2$ (stays inside ring) so it does travel in circle of radius $R/2$ about the cm. $T = 2\pi(R/2)/(v/2) = 2\pi R/v$.

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- They are different spring constant k's. Remember for springs in series: $\frac{1}{k_{\text{eff}}} = \frac{1}{k_1} + \frac{1}{k_2} + \cdots$. k_B is tricky but it doubles that force value because pulling x extends spring by $2x$ and it doubles that value again because the forces add on both sides (and they are the same force).

$$AM \geq GM \geq HM \quad \frac{a+b}{2} > \sqrt{ab} > \frac{2ab}{a+b}$$

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- Essentially, is it more likely for the ball to hit the plate when it is moving up or down? The answer is up because let's say the ball reached the collision zone's border. If the plate was moving up at any location, the ball will collide while plate moves up, so it will increase its speed. However, if plate is moving down and is at a low enough region, it has time to turn around and collide with the ball while moving up.